

Skin effect in a copper rod at 393 Hz

Hybrid $A-\Phi$ finite-element solver, tree-cotree gauge, validated against the **exact Bessel solution** for current density — magnitude *and* phase — and against **Kelvin’s exact AC resistance**.

regression_tests/04_Cylinder_SkinDepth · $a/\delta = 3.000$ · 418,318
unknowns · backward error 5.5e-21

0. What this case establishes

`01_OneCylinder` runs the same rod at 50 Hz, where $a/\delta = 1.07$ and $R_{ac}/R_{dc} = 1.027$. The skin effect is real there but it is only **2.6 %** of the answer — almost all of R is just R_{dc} , which any solver that integrates σ correctly will get right. Matching Kelvin to 0.038 % there validates *the effect itself* to only about 1.4 %.

This case runs the same rod at **393 Hz**, where the skin effect is **77 % of R** rather than a correction to it.

	01 @ 50 HZ	04 @ 393 HZ
a/δ	1.070	3.000
$ J(0)/J(a) $ exact	0.9265	0.2522 — 4x surface to axis
phase rotation, axis to surface	27.6°	148° — nearly anti-phase
R_{ac}/R_{dc} exact	1.0267	1.7680
the skin effect is	2.6 % of R	77 % of R
measured R_{ac}/R_{dc}	1.02633	1.77191
error against Kelvin	−0.038 %	+0.220 %
so the <i>effect</i> is validated to	~1.4 %	~0.29 %

Why the larger raw error is the stronger result. 0.038 % looks better than 0.220 % until you ask what fraction of the answer the skin effect actually is. At $a/\delta = 1.07$ the quantity under test is a 2.6 % correction to a number the solver would get right anyway. At $a/\delta = 3$ it is most of the answer, and $|J|$ falling 4x from surface to axis is something a wrong answer cannot fake.

1. The case

Conductor	copper, $\sigma = 5.8e7 \text{ S/m}$, $a = 10 \text{ mm}$, $L = 40 \text{ mm}$, a 76-gon
Domain	200 mm box of air, $\mathbf{n} \times \mathbf{A} = \mathbf{0}$ on the outer boundary
Excitation	1.0 V across the end caps, 393 Hz
Skin depth	$\delta = 3.3336 \text{ mm}$, so $a/\delta = 2.9998$
Formulation	first-order Whitney edge $\mathbf{A}$, second-order P2 nodal Φ
Mesh	31,922 nodes, 179,477 tets, 98,204 of them in the wire; Netgen-optimised
A_{poly}	313.8015 mm ² (the circle would be 314.1593)
R_{dc}	$L/(\sigma A_{\text{poly}}) = 2.197743e-06 \Omega$
Measured	$ I = 19977.92 \text{ A}$, $Z = 3.894203e-06 + 4.990355e-05 j \Omega$

Why 393 Hz and not more. At $a/\delta = 5$ (1092 Hz) the exact $|J(0)/J(a)|$ is 0.054, so the core current is 5 % of the surface value and a relative error there is being taken against nearly zero — the reference stops being well conditioned. It would also need $N = 128$ and about 1.7 M unknowns. $a/\delta = 3$ is the most demanding point that still has a meaningful exact reference.

2. The analytical solution

2.1 From Maxwell's equations to Bessel's equation

Inside the conductor displacement current is negligible ($\sigma \gg \omega\epsilon$), so with the $e^{j\omega t}$ convention:

$$\nabla \times \mathbf{H} = \mathbf{J} = \sigma \mathbf{E} \quad \nabla \times \mathbf{E} = -j\omega\mu\mathbf{H}$$

Taking the curl of the first, substituting the second, and using $\nabla \cdot \mathbf{B} = 0$ with constant μ :

$$\nabla^2 \mathbf{J} = j\omega\mu\sigma \mathbf{J}$$

Define the skin depth $\delta = \sqrt{2/\omega\mu\sigma}$. Then $j\omega\mu\sigma = 2j/\delta^2$, and writing this as a Helmholtz equation $\nabla^2 \mathbf{J} + k^2 \mathbf{J} = 0$ gives

$$k^2 = -j\omega\mu\sigma = -2j/\delta^2 \quad \rightarrow \quad k = (1 - j)/\delta$$

For a long cylinder the current is axial and depends only on r , so the Laplacian reduces to

$$(1/r) \frac{d}{dr} (r \frac{dJ_z}{dr}) + k^2 J_z = 0$$

which is **Bessel's equation of order zero**. Its two solutions are $J_0(kr)$ and $Y_0(kr)$; Y_0 diverges on the axis, so physics keeps only the first:

$$J_z(r) = J_z(a) \cdot J_0(kr) / J_0(ka)$$

A notation collision worth naming. J_z is current density; J_0 is the Bessel function. They are unrelated and both are standard.

This is not an exponential. $e^{-(a-r)/\delta}$ is only the asymptotic form for $a \gg \delta$. At $a/\delta = 3$ the Bessel solution is what is correct, which is why the measured profile *flattens* toward the axis rather than decaying to zero.

2.2 Kelvin functions: why the argument is complex

Because k is complex, $J_0(kr)$ is a Bessel function of complex argument. The **Kelvin functions** exist precisely to split that into real and imaginary parts:

$$\text{ber}(x) + j \cdot \text{bei}(x) \equiv J_0(x \cdot e^{3j\pi/4})$$

Since $kr = \sqrt{2} (r/\delta) e^{-j\pi/4}$ and J_0 is even, this is exactly $J_0(kr) = \text{ber}(u) + j \cdot \text{bei}(u)$ with

$$u = \sqrt{2} \cdot r/\delta \quad \text{and at the surface} \quad u = \sqrt{2} \cdot a/\delta = 4.242337$$

That is why $u = \sqrt{2} a/\delta$ appears throughout: it is the same Bessel function written in real arithmetic. The defining series are

$$\begin{aligned} \text{ber}(x) &= \sum (-1)^m (x/2)^{4m} / [(2m)!]^2 \\ \text{bei}(x) &= \sum (-1)^m (x/2)^{4m+2} / [(2m+1)!]^2 \end{aligned}$$

At this case's u :

VALUE AT $u = 4.242337$

$\text{ber}(u)$	-3.365444524
$\text{bei}(u)$	+2.095782021
$\text{ber}'(u)$	-3.475101853
$\text{bei}'(u)$	-1.160258139

2.3 The AC resistance — Kelvin's formula

Integrate the current density over the cross-section, using $\int_0^a J_0(kr) r dr = (a/k) J_1(ka)$:

$$I = \int_0^a J_z(r) \cdot 2\pi r dr = J_z(a) \cdot 2\pi a \cdot J_1(ka) / (k \cdot J_0(ka))$$

Divide the surface field $E_z(a) = J_z(a)/\sigma$ by that to get the internal impedance per unit length, and normalise by $R_{dc} = 1/(\sigma a^2)$:

$$Z_{\text{internal}} / R_{dc} = (ka/2) \cdot J_0(ka) / J_1(ka)$$

Taking the real part and rewriting in Kelvin functions gives the form used in `expected.txt` :

$$R_{ac}/R_{dc} = (u/2) \cdot [\text{ber}(u) \text{bei}'(u) - \text{bei}(u) \text{ber}'(u)] / [\text{ber}'(u)^2 + \text{bei}'(u)^2]$$

and the companion internal inductance, $\omega L_{\text{int}}/R_{dc} = (u/2) [\text{ber} \cdot \text{ber}' + \text{bei} \cdot \text{bei}'] / [\text{ber}'^2 + \text{bei}'^2]$, which evaluates to `1.463933` here.

2.4 Verifying the reference itself

Every error quoted below is measured *against* these formulas, so a bug in them would be invisible and would corrupt every number in the report. Three independent checks:

Two routes to the same quantity. The Kelvin form above and $\text{Re}[(ka/2) J_0(ka)/J_1(ka)]$ are the same number written differently, computed here through entirely separate code — real `ber/bei` series and derivative series versus complex `J0` and `J1` series:

```
(u/2)(ber bei' - bei ber')/(ber'^2 + bei'^2) = 1.768019136
Re[ (ka/2) J0(ka)/J1(ka) ] = 1.768019136
difference = 0.0e+00
```

They agree to machine precision at every a/δ from 0.1 to 5.

The Kelvin functions against their defining identity. The real `ber/bei` series was checked against `J0(x ej3π/4)` computed from the complex series — a different code path — and they agree to machine precision at every `x` tested up to 8, including this case's 4.242.

Both asymptotic limits.

A/Δ	COMPUTED	LIMIT
0.10	1.00000208	$1 + (a/\delta)^4/48 = 1.00000208$
0.50	1.00130073	1.00130208
10.0	5.25930	$a/(2\delta) + 1/4 = 5.25000$
30.0	15.25312	15.25000

That low-frequency limit also explains why cases 02 and 03 see nothing: at $a/\delta = 0.16$, $(a/\delta)^4/48 = 1.4e-05$.

A caution the series carries. The alternating `ber/bei` series suffers catastrophic cancellation at large argument — it returns nonsense by $a/\delta = 100$. It is exact to machine precision across the range used here ($u \leq 4.25$), but it is not a general-purpose implementation and should not be reused blindly at higher frequency.

3. The mesh, and how it was sized

3.1 The sizing was measured, not assumed

`01_OneCylinder`'s mesh was run at three frequencies first and the error read off against Bessel, to find what resolution the profile actually needs:

F	Δ	A/Δ	Δ/H	WORST BAND ERROR	R VS KELVIN
50 Hz	9.346 mm	1.070	5.66	0.26 %	-0.038 %
175 Hz	4.996 mm	2.002	3.03	2.10 %	-0.050 %
300 Hz	3.815 mm	2.621	2.31	4.73 %	+0.386 %

So the error is governed by δ/h , and $\delta/h \geq 3$ is what holds it near 1 %. At 393 Hz, $\delta = 3.334 \text{ mm}$, so h must be about 0.8–1.1 mm.

3.2 Where the refinement had to go — the non-obvious part

Reading the per-band error gets this backwards. At 300 Hz the error is 0.1 % in the annulus ($r > 0.8a$) and 4.7 % in the core ($r < 0.3a$), which looks like “refine the core”. It is not.

$|J|$ in the core is **flat** there — 0.3934, 0.3929, 0.3949 across r/a 0 to 0.3 — so the core is not failing to resolve local variation. It is **inheriting accumulated error** from the region where the field actually decays, and δ/h was 1.45 to 2.31 across the *whole* conductor. The annulus’s own relative error only looks small because $|J|$ is large there.

At $a/\delta = 3$ the decay region $r > a - 2\delta = 3.33 \text{ mm}$ is **89 % of the cross-section**, so there is nothing to gain by grading inside the conductor: $lc_{\text{core}} = lc_{\text{skin}}$ and the wire is meshed uniformly.

3.3 The facet rule fixes N

The polygon facet is $2a \sin(\pi/N)$, and the design rule is that the volume element size stays *comparable* to the facet width; a volume size well below the facet produces slivers. At $N = 36$ the facet is 1.743 mm, so $h = 0.833 \text{ mm}$ would be less than half of it. $N = 76$ gives **0.827 mm**, matched to h .

This is the same rule that killed an $N = 96$ experiment in case 02 from the other direction, where the facet was 0.098 mm and the volume size 0.25 mm. **The ratio is what matters, in either direction.**

3.4 The unsigned-distance trap

The mesh is graded by a **Distance** field measured from the conductor’s lateral surface, mapped through a **Threshold**. **That distance is unsigned**, so it reads 10 mm on the axis — the same as a point 10 mm out in the air — and the field coarsens *inward* as well as

outward. In `01_OneCylinder` that left the core at 4.4 mm elements with **no cells at all inside** `r < 0.1a` .

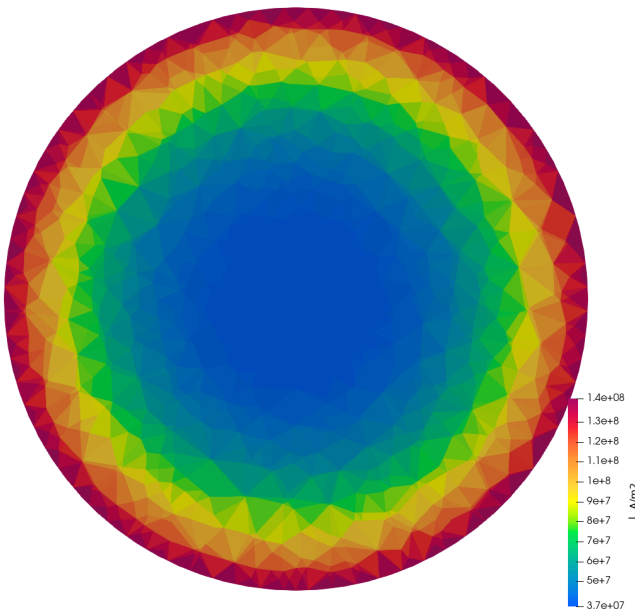
The fix, used here and in case 02, is a second field capping the interior:

```
Field[3] = MathEval;   Field[3].F = Sprintf("%g", lc_core);  
Field[4] = Restrict;   Field[4].InField = 3;   VolumesList = {wire_volume};  
Field[5] = Min;        Field[5].FieldsList = {2, 4};
```

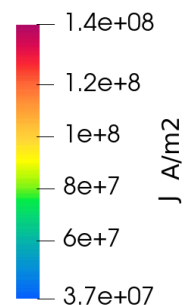
`Restrict` returns a huge size outside its volume, which is exactly what `Min` wants, so `lc_core` binds only inside the wire and the air grading is untouched. A third field holds the fine size through the first half skin depth outward (`DistMin =  $\delta/2$` rather than 0) — without that, the fine size decays immediately and there is a fine *surface* but no resolved *shell*.

4. Results

4.1 The picture



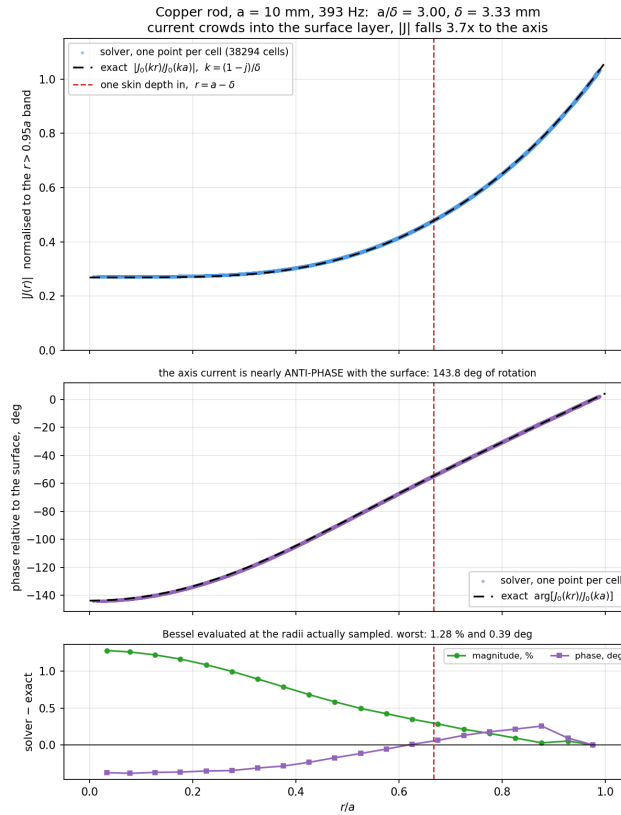
|J| at mid height. The current is confined to a surface layer about one skin depth thick; the core carries almost none of it.



Longitudinal slice. Nothing varies along `z` , which is what the infinite-cylinder assumption behind the analytical solution requires.

Unlike the 50 Hz cases, the auto colour range here is *meaningful*: $|J|$ spans nearly 4x across the section, so the visible structure is physics rather than noise in the fifth digit.

4.2 Magnitude and phase against the exact Bessel solution



38,294 cells, one point each, against $J_0(kr)/J_0(ka)$ evaluated at the radii actually sampled. Top: magnitude. Middle: phase. Bottom: both errors.

There is no radial probe line. Every point is one tetrahedron plotted at its centroid radius, over **all azimuths** and all z in $0.3L$ to $0.7L$. A line would have allowed a lucky azimuth to be chosen and would have sampled about 30 cells instead of 38,294. The vertical spread at fixed r is the azimuthal scatter and element noise, shown rather than averaged away. The values are the per-cell J , exact per tetrahedron, not the volume-averaged nodal field.

Both curves are normalised by the *same* reference — the mean over cells with $r > 0.95a$ — so amplitude and the drive's own phase cancel and only shape is compared. Amplitude is checked separately by R , which is an integral of the same field.

R/A	CELLS	J MEASURED	J EXACT	ERR	PHASE ERR
0.00–0.10	334	0.2717	0.2683	+1.27 %	−0.385°
0.10–0.20	958	0.2724	0.2692	+1.19 %	−0.372°
0.20–0.30	1576	0.2770	0.2741	+1.04 %	−0.353°
0.30–0.40	2248	0.2917	0.2893	+0.84 %	−0.302°
0.40–0.50	2842	0.3236	0.3216	+0.63 %	−0.206°
0.50–0.60	3323	0.3798	0.3781	+0.46 %	−0.086°
0.60–0.70	3900	0.4651	0.4636	+0.31 %	+0.036°
0.70–0.80	4413	0.5841	0.5831	+0.18 %	+0.156°
0.80–0.90	5212	0.7458	0.7454	+0.05 %	+0.238°
0.90–0.95	3455	0.8890	0.8886	+0.05 %	+0.092°
0.95–1.00	10033	1.0000	1.0000	0.00 %	0.000°

Worst band: 1.27 % in magnitude, 0.385° in phase, and both are *monotonic* — largest at the axis, vanishing at the surface. That shape is the signature of an h -limited solution rather than a wrong one: the error is largest exactly where δ/h is worst and disappears where the mesh is finest.

4.3 Phase is the tighter test

J_z is a phasor and the magnitude comparison tests only half of it. The argument is independent, and in practice more demanding: an error in the $-j\omega A$ term of $E = -j\omega A - \nabla\Phi$ shows up in the phase before it shows up in $|J|$. The exact rotation across the radius:

R/A	0.0	0.2	0.4	0.6	0.8	1.0
$ J_o(kr)/J_o(ka) $	0.2522	0.2543	0.2833	0.3895	0.6119	1.0000
phase, deg	−148.09	−137.81	−108.92	−71.42	−34.91	0.00

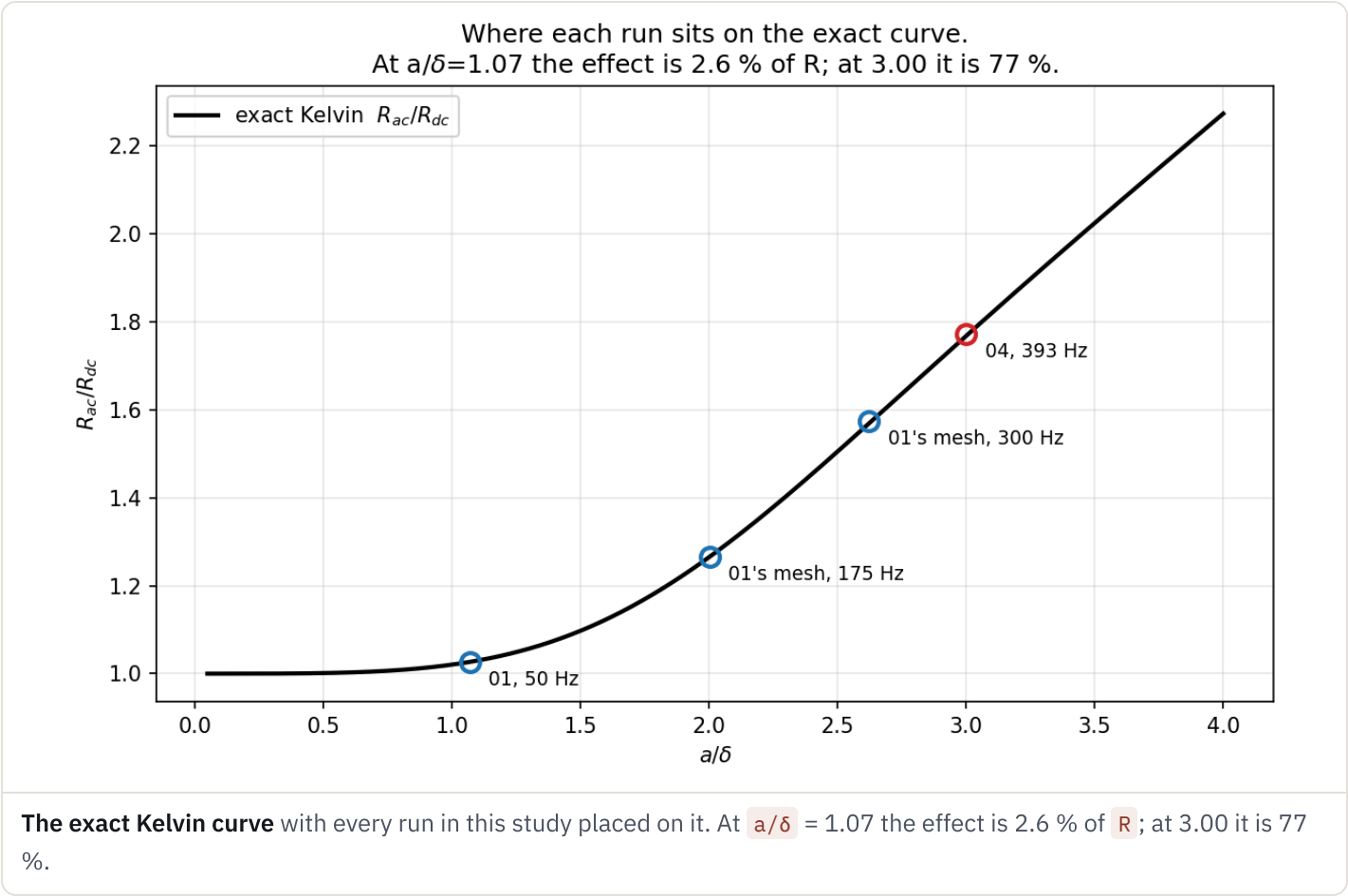
The axis current is **nearly anti-phase** with the surface. Across the whole suite:

CASE	A/ Δ	MEASURED LAG	EXACT	ERROR	(MAGNITUDE ERROR)
02 / 03	0.16	-0.62816°	-0.63860°	0.010°	0.00 %
01	1.07	-27.4249°	-27.5592°	0.134°	+0.26 %
01's mesh @ 300 Hz	2.62	-112.673°	-111.979°	0.694°	+4.7 %
04	3.00	-134.239°	-133.847°	0.392°	+1.15 %

This case reproduces a **134° rotation to 0.39°** — 0.29 % of the rotation, *tighter* than the 1.15 % on the magnitude ratio. Case 02 is the most telling of the others: its magnitude ratio is 0.99997, flat to five digits and nearly insensitive, yet it resolves a 0.64° lag to 0.010°.

Two details that had to be right. The **complex field is averaged and the argument taken afterwards**; averaging the arguments is wrong wherever the phase spread inside a band is not small, which is exactly the high-frequency case. And everything is referred to the same $r > 0.95a$ band, so the drive's own phase cancels — ϕ is gauge dependent, but $J = \sigma E$ is not, so the result is physical rather than a convention.

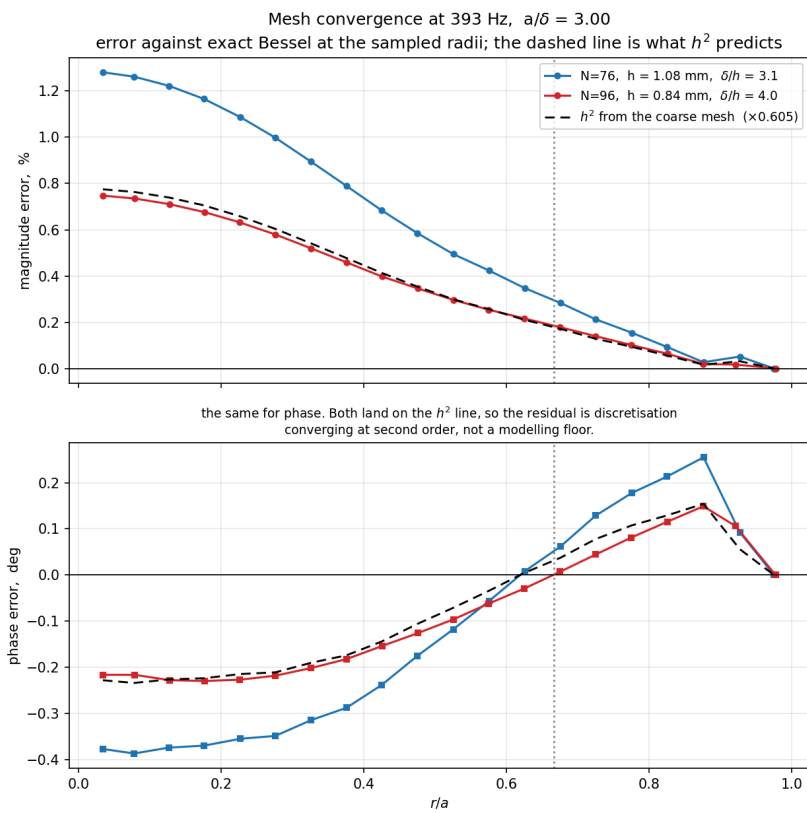
4.4 AC resistance against Kelvin



Kelvin exact	1.768019
measured	1.771910
error	+0.220 %
of which the polygon accounts for	+0.028 % — see §7
L from $\text{Im}(Z)/\omega$	20.2097 nH
$\omega L/R$	12.8 — strongly inductive

5. Mesh convergence: N=76 against N=96

`cylinder_skin_n96.geo` is the same case at $N = 96$, $l_c = 0.640 \text{ mm}$, which reaches $\delta/h = 4.0$ in the conductor. It is **not** a regression case — 739,304 unknowns and 17.1 GB of factors is too heavy to run on every change — it exists to establish the convergence *rate*.



The dashed line is what h^2 predicts from the coarse mesh. The fine mesh lands on it at every radius, in magnitude and phase alike.

	N=76	N=96	RATIO	IMPLIED ORDER
h in the conductor	1.08 mm	0.84 mm	0.778	
δ/h	3.1	4.0		
worst band $ J $	1.27 %	0.74 %	0.58	2.17
worst band phase	0.385°	0.230°	0.60	2.06
R vs Kelvin	+0.220 %	+0.145 %	0.66	1.65
L	20.2097 nH	20.2041 nH		converged to 0.03 %
unknowns	418,318	739,304	1.77	
factor nnz	497.8 M	1068.6 M	2.15	
factorization	25.9 s	81.8 s	3.15	

The rate matters more than either error value. A single number says “we are 0.74 % off” and cannot distinguish discretisation error from a modelling mistake sitting at a floor. Two meshes give the rate: h falls by 0.778 and the error falls by $0.605 = 0.778^2$, at every radius, in magnitude and phase alike. So the residual is genuine discretisation converging at **second order**, and refinement would keep paying. An error that had stalled between the two meshes would have meant the opposite, and would have pointed at the geometry or the formulation rather than at h .

R converges more slowly (order 1.65) because part of its error does not scale with h at all: the polygon floor, which fell 0.028 % to 0.018 % only because N changed. Net of it, 0.192 % to 0.127 %.

6. Cost

nodes / tets	31,922 / 179,477 (98,204 in the wire)
unknowns	418,318 — A : 176,554 free edges, 28,996 gauged, 8,772 Dirichlet
factor nnz	497.8 M, about 8.0 GB
mesh	24 s
factorization, MUMPS + threaded MKL	25.9 s
one frequency end to end	42 s
backward error	5.5e-21

This is the first case that cannot practically run on the internal solver, so its

`expected.txt` names the MUMPS input — the opposite of cases 01–03, which name their plain inputs on purpose so `check.bat` exercises the default backend.

Measured scaling on this geometry, across case 01’s mesh and both meshes here: $fill \sim n^{1.47}$, factorization time $\sim n^{2.39}$ on a sequential BLAS. Linking MKL’s threaded layer (`-DMKL_THREADING=intel_thread`) is worth about **10x** on this problem — 266.78 s to 25.94 s — because nearly all the flops are dense kernels inside the frontal matrices, and a sequential BLAS runs those on one core. See `docs/MUMPS_SETUP.md`.

7. Limitations, and what “exact” does not mean

The analytical solution is exact for an **idealisation**: an infinitely long, perfectly circular conductor of uniform σ and μ , quasi-static. The simulation differs from it in three ways, and the residual error is discretisation *plus* those differences, not discretisation alone.

- **The conductor is a 76-gon, not a circle.** At $a/\delta = 3$ the current lives in a thin surface layer, so what matters is the *perimeter* rather than the area: 62.8140 mm against the circle’s 62.8319 mm, or -0.0285% , which raises R_{ac} by about **+0.028 %**. **This will not go away with a finer mesh** — only with a larger N , which also changes R_{dc} . A tolerance tight enough to exclude it would be asserting something false.
- **The rod is 4 radii long, not infinite.** Mitigated by sampling only the mid-length band $0.3L$ to $0.7L$, and the longitudinal slice shows no z dependence there.
- **The air box is finite**, with $n \times A = 0$ at 200 mm, against the analytical solution’s unbounded exterior.

Φ is gauge dominated here and is deliberately not checked. At $\omega L/R = 12.8$ the reactance dominates, and under the tree-cotree gauge Φ differs from the z/L gradient by $-j\omega\psi$, which is the larger term; measured, Φ/V deviates from z/L by 0.51. That was established by experiment rather than assumed: running `01_OneCylinder`’s *same mesh* at 1 Hz instead of 50 Hz — identical elements, only ω changed — moves the deviation from $5.4e-01$ to $1.9e-03$, a factor of 253. Discretisation cannot do that. E , B , H and J are gauge independent and are what this case validates.

Memory, not time, is what stops further refinement on a 32 GB machine. Halving h again needs $N \sim 136$ and about 1.8 M unknowns, which extrapolates to roughly 45 GB of factors. Going materially below 0.5 % needs a different lever — second-order elements for A , or an out-of-core solve — not a smaller h .

8. What is asserted automatically

`regression_tests/check.bat` runs this case and asserts all five, against the values in `expected.txt`:

V measured vs prescribed	1.000000 vs 1.0
R/R_dc vs Kelvin	1.77191 vs 1.768019, tol 0.006
L	20.2097 nH, tol 0.05 — pinned to catch change
<J>core/<J>surf	0.281160 vs 0.277956, tol 0.005
phase lag, core to surface	−134.239° vs −133.847°, tol 0.8°

The two Φ checks are deliberately *absent*, and `verify.py` prints `SKIPPED` with the reason rather than passing silently. An omitted key is a claim about the physics, not an exemption.

Reproducing

```
gmsht cylinder_skin.geo -3 -o cylinder_skin.msh
..\run_case.bat cylinder_393hz_mumps.aphi
"C:\Program Files\ParaView 6.1.1\bin\pvbatch.exe" make_plots.py output
sh build_doc.sh # rebuilds this document

# the convergence study
gmsht cylinder_skin_n96.geo -3 -o cylinder_skin_n96.msh
..\run_case.bat cylinder_393hz_n96_mumps.aphi
"C:\Program Files\ParaView 6.1.1\bin\pvbatch.exe" make_convergence_plot.py
```

Run `build_doc.sh` after regenerating figures, or this document keeps showing the previous run’s pictures — they are baked into the HTML and the PDF.